

Reflection Journal Report for Midterm

Prepared for Prof. Vishwantha Rao, HCC ITAI 1371 Machine Learning

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The purpose of this report is for each student to reflect on their learning journey for ITAI 1371 Machine Learning (ML) MidTerm. This Reflection Journal Report consists of separate parts for each student in the group. Each student in the group was responsible for their own content and report formatting in each of the sections below.

This reflection journal entry is for the Midterm Exam was prepared by Eric Roberson.

Understanding Insights & New Breakthroughs

Analyzing the Problem / Problem Solving Strategies

Real-World Applications

Connecting New & Prior Knowledge

Emerging Questions

Changing Approach & Critical Thinking

The midterm asked us to respond to the following critical thinking questions:

This reflection journal entry is for the MidTerm Exam and was prepared by Irene(Alex) Ramos:

Understanding Insights & New Breakthroughs

While developing our midterm proposal, I gained a clearer understanding of how exploratory data analysis prepares a dataset for machine learning. Before this project, I mostly thought of data cleaning as removing empty information or correcting obvious errors. I now understand that preprocessing decisions can directly affect the accuracy and fairness of a future model. One of my biggest breakthroughs was learning that the original rating column must be removed from the model inputs after it is used to create the “kid_safe” target. Keeping it would cause target leakage because the model would essentially be given the answer it is supposed to predict.

I also learned why the training and testing data should be separated before completing EDA. We plan to use 70% of the data for training and reserve 30% for testing. Analyzing the training set while leaving the test set untouched will give us a more honest evaluation of the model later.

Analyzing the Problem / Problem-Solving Strategies

Our project asks whether the characteristics of a Netflix movie can help classify it as potentially appropriate for a children’s profile. To make the problem manageable, we decided to filter the original Netflix dataset so that it contains only movies rather than both movies and television shows. We will create a binary target called “kid_safe”, with G- and PG-rated movies coded as 1 and clearly mature ratings coded as 0. Movies with missing or unclear ratings will be separated for review instead of being automatically labeled.

Our problem-solving strategy is to complete each step through reproducible Python code. We will inspect the dataset for duplicates, missing values, incorrect data types, skewed numerical distributions, and class imbalance. Instead of applying the same solution to every missing value, we will choose an appropriate method for each feature. For example, some missing categorical values may be assigned an “Unknown” category, while records missing

information needed to create the target may need to be excluded from the modeling dataset.

Real-World Applications

This project has practical applications for streaming platforms, parents, and families. A classification model could help identify movies that may be suitable for a children's profile, especially when rating information is incomplete, inconsistent, or unavailable. Features such as genre, description, duration, release year, country, director, and cast may contain patterns that help support this decision.

However, I also recognize that a machine learning prediction should not replace parental judgment or an official content rating. A movie may have a G or PG rating and still contain themes that some families consider inappropriate. Therefore, the model would work best as a recommendation or screening tool rather than as the final authority on whether a child should watch a movie.

Connecting New & Prior Knowledge

This proposal helped me connect concepts from earlier coursework to a complete machine learning workflow. I had previously worked with pandas, missing values, categorical data, and visualizations, but this project helped me understand how those skills work together. Loading the original dataset, inspecting its structure, filtering records, engineering new features, encoding categories, and visualizing distributions are not separate activities. Each step contributes to creating reliable data for a future model.

I also connected what I know about classification to the importance of selecting a clear target variable. Changing our original idea from predicting several content-rating categories to a binary "kid_safe" classification made the problem more focused. It also helped us define which rows belong in the project and which features might be useful predictors.

Emerging Questions

As we continue the project, several questions remain. How should ratings such as TV-PG or TV-Y7 be handled if they appear on records labeled as movies? Should PG-13 movies automatically be considered unsafe, or should they form a separate category for human review? How many G- and PG-rated movies will remain after filtering, and will that create serious class imbalance?

I am also interested in finding out which engineered features will be the most useful. For example, will keywords from the description provide stronger information than genre or

duration? How should columns containing multiple values, such as cast, country, and listed genres, be encoded without producing too many features? Another question is whether the definition of “kid-safe” is too subjective to be represented by one binary target.

Changing Approach & Critical Thinking

Our approach became more specific as we reviewed the assignment requirements and thought more carefully about the data. Instead of manually editing the original dataset, we decided that every transformation should be completed through Python so that the process can be repeated and verified. We also changed our approach by separating movies from television shows, defining a clear binary target, and planning to reserve an untouched test set.

The most important change in my thinking was realizing that preprocessing methods should be chosen only after examining the data. We should not automatically scale every numerical variable, remove every row with a missing value, or balance the classes without evidence that those actions are necessary. We will first study the training data and then document the reasoning behind each decision. This project has encouraged me to think more critically about how data choices influence the final model and to focus on building a process that is transparent, reproducible, and appropriate for the problem.

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